

ANSWERS

1.

Question	Answer	Marks
2(a)	change / increase of velocity (with time)	B1
	change / increase of velocity per unit time or rate of change / increase of velocity	B1
2(b)(i)	force $\times$ distance moved <u>in the direction of the force</u> / <u>forwards</u>	B1
2(b)(ii)	(work done $\Rightarrow$) $5.4 \times 10^4 \times 280$	C1
	1.5×10^7 (J)	A1
2(b)(iii)	$E_k = \frac{1}{2}mv^2$ or ($m \Rightarrow$) $2 \times E_k / v^2$	C1
	($m \Rightarrow$) $2 \times 3.2 \times 10^6 / 12^2$ or $2 \times 3.2 \times 10^6 / 144$	C1
	4.4×10^4 (kg)	A1

2.

Question	Answer	Marks
2(a)(i)	chemical (potential energy)	B1
2(a)(ii)	<u>electrical</u> work / (electrical) current or electrons (made to) move	B1
	mechanical) <u>work</u> is done by the <u>motor</u> / by or on the (back) <u>wheel</u>	B1
2(b)	(KE $\Rightarrow$) $\frac{1}{2}mv^2$ in any form	C1
	560 (J)	A1
2(c)(i)	(energy provided if power is) 1 kW for 1 hour / 1000 W for 1 hour	B1
2(c)(ii)	($t \Rightarrow$) E/P or $0.35 / 0.07$ or $350 / 70$	C1
	5(.0 hours)	A1

3.

Question	Answer	Marks
2(a)	(change in g.p.e.) mgh or $0.12 \times 10 \times 0.45$	C1
	0.54 J	A1
2(b)	k.e. $= \frac{1}{2}mv^2$ or $0.54 = \frac{1}{2} \times 0.12 \times v^2$	C1
	($v \Rightarrow$) $\sqrt{2gh}$ or $\sqrt{2 \times 10 \times 0.45}$ or $v^2 = 2 \times 0.54 / 0.12$ or $v^2 = 9$	C1
	3.0 m / s	A1
2(c)(i)	not enough (kinetic) energy (at C) to travel beyond D / to top of circle or more g.p.e. needed originally to travel beyond D / to top of circle or g.p.e. gained cannot be greater than original g.p.e. (transferred to k.e)	B1
2(c)(ii)	it travels back down to C and returns to A	B1

4.

Question	Answer	Marks
2(a)	$(\rho =) m / V$ or $(\rho =) m / hA$ or $2.9 \times 10^4 / 3.6$ or $2.9 \times 10^4 / (2.0 \times 1.8)$	C1
	$8.1 \times 10^3 \text{ kg / m}^3$	A1
2(b)(i)	(g.p.e. =) $mg\Delta h$ or $2.9 \times 10^4 \times 10 \times 0.80$	C1
	$2.3 \times 10^5 \text{ J}$	A1
2(b)(ii)	k.e. / $\frac{1}{2}mv^2 = \text{g.p.e.} / mg\Delta h / 2.3 \times 10^5$ or $(v =) \sqrt{2gDh}$ or $\sqrt{2 \times 10 \times 0.80}$	C1
	$(v =) \sqrt{2gDh}$ or $\sqrt{2 \times 10 \times 0.80}$	C1
	4.0 m / s	A1

5.

Question	Answer	Marks
2(a)	(K.E. =) $\frac{1}{2}mv^2$ numerical or algebraic	C1
	$0.5 \times 60\,000 \times 12^2$	C1
	4 300 000 J	A1
2(b)(i)	reduces global warming	B1
2(b)(ii)	wind not always blowing or wind not reliable / not available in some places or smaller land area (for same power) or less harm to birds or less visual effect or no interference with tv / radar signals	B1
2(c)(i)	(amount of energy of a device of power) 1 kW used for 1 hour	B1
2(c)(ii)	23 000 kg	B1
2(d)	solar, tidal, geothermal, hydro(electric), biomass, wave	B1

6.

Question	Answer	Marks
4(a)	chemical (energy)	B1
4(b)(i)	energy cannot be created and energy cannot be destroyed	B1
	energy can be transferred from one form into another	B1
4(b)(ii)	some chemical energy is transferred to thermal energy (sound and light)	B1
	thermal energy lost to air / water / buildings / equipment	B1
4(c)	any one from: carbon dioxide produced; carbon monoxide produced; global warming; greenhouse gases nitrous oxide produced; sulfur dioxide produced; acid rain; rise in sea-level	B1

7.

Question	Answer	Marks
9(a)(i)	chemical	B1
9(a)(ii)	electrical	B1
9(a)(iii)	ANY 2 from - light - sound - radio / microwave / electromagnetic (waves)	B2
9(b)	(Q =) It in any form	C1
	0.16 C	A1
9(c)(i)	($E =$) mcT numerical or algebraic in any form	C1
	$0.11 \times 830 \times 5$	C1
	460 J	A1
9(c)(ii)	energy output / energy input	C1
	useful energy output / (total) energy input	A1
9(c)(iii)	(energy input =) 460 + 5200 or 5660 (J) seen or 5200 + (energy in 9(c)(i) + 5200)	C1
	92% or 0.92	A1
9(c)(iv)	(thermal) energy <u>lost</u> / heat <u>lost</u> (from battery to environment)	B1
	energy input larger than calculated / 5660 (J) or temperature rise should be larger / is smaller than it should be or thermal energy / heat produced larger than 460 J / calculated	B1

8.

8(c)(i)	K.E. = $\frac{1}{2}mv^2$ in any form algebraic or numerical	C1
	$v^2 = 2 \times 4000 / 520$ or $v = \sqrt{\frac{2 \times 4000}{520}}$	C1
	3.9 m / s	A1
8(c)(ii)	($h =$) P.E. / mg in any form algebraic or numerical	C1
	$3000 / (10 \times 520)$ or 0.58 m	C1
	2.0 m	A1

9.

Question	Answer	Marks
2(a)(i)	(g.p.e. =) mgh or $90 \times 10 \times 0.60$	C1
	540 J	A1
2(a)(ii)	(W.D. =) Fx or 290×2.0	C1
	580 J	A1
2(a)(ii)	friction (between wheel and axle) or friction (between tyre / wheel and plank) or (work done against) friction	B1
2(b)(i)	$\frac{\text{useful energy output}}{\text{(total) energy input}} (\times 100\%)$ or $\frac{\text{useful power output}}{\text{(total) power input}} (\times 100\%)$	B1
2(b)(ii)	wheelbarrow / worker has to be lifted up ignore references to friction / air resistance	B1

10.

Question	Answer	Marks
6(a)	worn insulation → visual check loose wire in live → earth wire cable too hot → circuit breaker ALL three correct B2 one or two correct B1	B2
6(b)(i)	energy used by (a device of power) 1 kW for 1 hour	B1
6(b)(ii)	(number of kW h =) $1.2 \times 20 / 60$ or 0.4 seen	C1
	8 c	A1

11.

Question	Answer	Marks
10(a)(i)	ammeter in series connecting switch with motor	B1
	voltmeter across motor or battery	B1
10(a)(ii)	(E =) VI in any form numerical or algebraic or $P = VI$ and $P = E / t$ in any form	C1
	48 J	A1
10(a)(iii)	(E = mgh) in any form numerical or algebraic	C1
	$0.15 \times 0.8(0) \times 10$	C1
	1.2 J	A1
10(a)(iv)	ANY two of - motor / mass has kinetic energy - heat generated in motor / wires - heat generated in air due to drag / resistance <u>work done</u> to lift string	B2

12.

Question	Answer	Marks
2(a)	mgh and $\frac{1}{2}mv^2$	B1
2(b)	$\frac{1}{2}mv^2 = mgh$ or $v^2 = 2gh$ or $v^2 = 2 \times 10 \times 380$ or $v^2 = 7600$	C1
	$(v =) \sqrt{2gh}$ or $\sqrt{2 \times 10 \times 380}$	C1
	87 m / s	A1
2(c)(i)	both the g.p.e. and the k.e. are proportional to mass or mass cancels out in the equation or they accelerate at the same rate / g	B1
2(c)(ii)	air resistance (force) is a smaller proportion of weight (for heavier coin)	B1
	acceleration of heavier coin greater / less affected by air resistance	B1

13.

Question	Answer	Marks
3(a)	<i>in spring:</i> elastic or strain (potential energy)	B1
	<i>in muscles:</i> chemical (potential energy)	B1
3(b)	spring extends a different distance for the same (extra) load or load is no longer proportional to extension or becomes easier to pull	B1
3(c)(i)	(W =) Fd numerical or algebraic; 400×0.23	C1
	92 J	A1
3(c)(ii)	measure / determine the no. of times he extends the spring in 60 s	B1
	power = (total) work / time explained	B1

14.

- (a) (i) will not run out **or** infinite **or** being replaced **B1**
(ii) solar / Sun, tidal, geothermal, biomass, hydroelectric, **B1**
water waves
(iii) 1 (burning) use of a fossil fuel, e.g. coal, oil, natural gas etc. **B1**
2 produces carbon dioxide _____ **B1**
or greenhouse gases (absorb infra-red from Earth)
(b) (i) (E=) $\frac{1}{2}mv^2$ seen in any form algebraic or numerical **C1**
 $4.7 \times 10^9 \text{ J}$ **A1**
(ii) (E=)Pt seen in any form algebraic or numerical **C1**
 $1.5 \times 10^9 \text{ J}$ **A1**
(iii) (efficiency=) (energy) output / (energy) input in any form **C1**
0.32 **or** 32% **A1**

15.

- (a) force $\times$ distance (moved) **C1**
force $\times$ distance moved in direction of force **A1**
(b) mgh in any form numerical or algebraic **C1**
21 J **A1**
(c) larger force is needed (by woman / on rope) **B1**
(because of either) **B1**
 - friction / heat produced
 - weight of rope / her arms
 - flag / rope gains K.E.
 - flag accelerates / gains speed

- 16.**
- (a) use of stopwatch or electronic timer **B1**
time at least 5 swings **and** divide by number of swings **B1**
or use of fiducial mark
or definition of one swing clear e.g. A to C to A or from A and back to A
- (b) (i) $(m=)$ P.E. / gh **or** $240 / 10 \times 0.6$ **C1**
40 kg **A1**
- (ii) resistance **or** friction (with air or rope and tree) **B1**
heat produced / work done (in / against air or friction) **B1**
or effect of wind
or work done by arms / legs
- 17.**
- (a) (i) not being replaced **or** will run out **B1**
(ii) only oil in 1st column **B1**
only wind and hydroelectric in 2nd column **B1**
geothermal in 3rd column **B1**
- (iii) less greenhouse gases / global warming / acid rain / **B1**
or toxic gases / oil spills **and** how they affect a named organism / ecosystem
- (iv) nuclear waste / radioactive waste causes (storage) problems **or**
explosion / melt down / leak emits radioactivity **B1**
- (b) (i) generator (and transformer) **B1**
(ii) kinetic (energy) **B1**
- 18.**
- (a) chemical (potential energy) **B1**
- (b) (i) non-renewable **and** oil / it is not replaced / will run out **B1**
(ii) acid rain **or** produces CO₂ **or** warms lakes / rivers / sea **or** global warming **B1**
or greenhouse effect
- (c) (i) useful energy output / (total) energy input **or** power for energy twice **B1**
- (ii) **1** $1.9 \times 10^9 / 0.38$ **or** $1.9 \times 10^9 \times 100 / 38$ **C1**
 $5.0 \times 10^9 \text{ W}$ **A1**
- 2** $(E =)Pt$ **or** $0.62 \times 5.0 \times 10^9 \times 2.0 (\times 3600)$ **or** $(5.0 - 1.9) \times 10^9$ etc. **C1**
 $2.2 \times 10^{13} \text{ J}$ **A1**

19.		
(a)	(PE =) mgh or Fd or 5×3.5 17.5 J or 17 J or 18 J	C1 A1
(b) (i)	(efficiency = useful) energy output/energy input in any form but all three quantities must be mentioned if efficiency is not the subject of the equation	B1
(ii)	17.5/0.65 or 17.5/65 or 0.65/65 = (a)/energy input 26.9 J or 27 J	C1 A1
(c)	due to friction (in bearings of motor) or due to (electrical) resistance (in motor) or air resistance acts or thermal energy/heat produced/lost (in resistance of motor/due to friction)	B1
20.		
(a)	($P =$) VI or 230×27 6200/6210 W or 6.21/6.2 kW	C1 A1
(b) (i)	1.1/1.12/1.1178 $\times 10^7$ J	B1
(ii)	6.21 $\times 0.5 \times 23$ or 6.21 $\times 30 \times 23$ or 3.1/3.105 $\times 23$ 71/71.3/71.4/71.415 c or \$ 0.71/0.714/0.71415	C1 A1
21.		
(a)	($V =$) m/ρ or 10 600/6500/4100/2400 $\div$ 1000 or 6.5 or 4.1 2.4 m ³ /2.4 $\times 10^6$ cm ³	C1 A1
(b) (i)	fuel/chemical (potential energy)	B1
(ii)	some to heat/thermal (energy) some to kinetic (energy of air or tractor)	B1 B1
(c)	($GPE =$) mgh or $2400 \times 10 \times 850$ 2.0/2.04 $\times 10^7$ J	C1 A1
22.		
(a)	(k.e. =) $\frac{1}{2}mv^2$ $\frac{1}{2} \times 4.4 \times 10^4 \times 20^2$ 8.8 $\times 10^6$ J	C1 C1 A1
(b) (i)	$WD = F \times x$ (ii) or force $\times$ distance (parallel to/in direction of force)	B1
(ii)	8.8 $\times 10^6$ /40 or $a = (-)5.0$ (m/s ²) or $t = 4.0$ (s) seen 2.2 $\times 10^5$ N	C1 A1

23.

- (a) (i) mass is the amount of matter/substance **or** to resist (change in) motion **or** (measurement of) inertia B1
weight is the pull/force of gravity **or** pull of Earth B1
accept mg where g stated as 10 (N/kg) or grav. field strength or acc. due to gravity
ignore mg where g is gravity or grav. force or undefined
- (ii) mgh algebraic or numerical C1
1200 J A1
- (iii) $\frac{1}{2}mv^2$ algebraic or numerical C1
(speed =) 9/12 **or** 0.75 seen A1
5.6(25) J A1
- (iv) ($E=$) VIt algebraic or numerical C1
4100 J **or** 4140 J A1
- (v) 1. energy can neither be created or destroyed/lost B1
(but) may change form / be transferred B1
2. electrical energy changes to P.E. (and K.E. and heat/work against friction; **ign.** mechanical energy) B1
- (b) (i) will not run out **or** infinite **or** being replaced (**allow** does not finish/always available)
(**ign.** cannot be reused/recycled) B1
- (ii) wind, tidal, solar/Sun, geothermal, hydroelectric, biomass, waves, wood (**not** nuclear) B2
(**allow** biogas/biofuel e.g. cane into petrol, dung into gas etc.) (**ign.** tidal waves)

24.

- (a) (i) ($WD =$) mgh **or** $54 \times 10 \times 2.8$ C1
1500/1510/1512 J A1
- (ii) ($P =$) WD/t or E/t **or** 1500/3 **or** 1510/3 **or** 1512/3 C1
500/503/504 W A1
- (b) any **two** of:
also lifting board/rope
heat in motor/wires/cable
friction with something named e.g. axle/spindle/air B2

- 25.
- (a) (i) (efficiency =) useful energy ÷ input energy C1
or $95\,000/120\,000 (\times 100)$
0.79(17) or 79(.17)% A1
- (ii) ($P =$) energy/time or $90\,000/60$ C1
1500 W A1
- 26.
- (a) any **two** from:
biomass/wood; geothermal power; solar power; tidal power; wave power; wind power B2
- (b) (i) 1. $2.1(4) \times 10^{17}$ J (**allow** $2.1(5) \times 10^{17}$ J if candidate uses 365.24/5) B1
2. any **one** from: not enough water (to maintain maximum flow); rainfall varies (during the year); periods of low demand B1
- (ii) 1. (GPE =) mgh or $1.6 \times 10^{10} \times 10 \times 170$ C1
 $2.7(2) \times 10^{13}$ J A1
2. $2.7(2) \times 10^{13}/3600$ or $6.8 \times 10^9 \times 3600$ or $6.8 \times 10^9/7.5(55) \times 10^9$ or
 $2.4(48) \times 10^{13}/2.7(2) \times 10^{13}$ C1
0.90 or 90% A1
3. any **two** from:
friction (of water) with pipe/turbine/; viscosity of water; friction at bearings; resistance/heat in the wires; KE of water leaving turbine B2
- 27.
- (a) (i) (PE =) mgh or $75 \times 10 \times 20$ C1
 1.5×10^4 J A1
- (ii) $\frac{1}{2}mv^2$ or $\frac{1}{2}75v^2$ C1
 $v^2 = 400$ (if this is seen it scores the first 2 marks) C1
 $v = 20$ m/s A1
- (b) (G)PE at start B1
KE at start B1
to elastic/strain/clear equivalent /EPE at end
(not stretch energy; any intermediate energy –1) B1
- 28.
- (a) (K.E.) = $\frac{1}{2}mv^2$; $\frac{1}{2} \times 90 \times 5^2$ C1
1125 J or 1100 J A1
- (b) (i) no resultant force; forwards force = backwards force/drag/friction (ignore air resistance) B1
water resistance/**water** drag mentioned/**water** friction OR sail exerts force on board B1
- (ii) heat produced OR equal to work done against backwards force/drag/friction B1

29.		
(a)	(i) 7000 (J) seen or 50% used somewhere ($P = E/t$) // 14000/t // 7000/t seen // 7000 J/minute // 420 000 J/hour 120 W	C1 C1 A1
	(ii) water after hitting turbine still moves // has KE/energy/velocity // energy lost due to friction // friction and location // heat/internal energy and location // water misses turbine	B1
(b)	(i) can be replaced/made // will not run out	B1
	(ii) coal, oil, gas, peat, nuclear, uranium (not solar)	B1
30.		
(a)	energy cannot be created/destroyed (nb. only one required) energy cannot be destroyed or created (i.e. the other one as well) or (merely) transformed or total energy in an isolated system is constant	B1 B1
(b)	(i) chemical (potential) at beginning to electrical (and heat) at end others present: max 1	B1 B1
	(ii) light heat/thermal/internal others present: max 1	B1 B1
(c)	less heat; same light or less chemical/electrical; less heat or less chemical/electrical; same light	B2 B2 B2
31.		
(a)	energy/time 1 joule in one second	C1 A1
(b)	(i) 5800 N or 5684 N or 5700 N	B1
	(ii) mgh algebraic, words or numerical (i.e. $580 \times 10 \times 12$) 69600 J or 70000 J or 68208 J or 68000 J	C1 A1
	(iii) (efficiency =) output power or energy/input power or energy algebraic or numerical or 93000 seen or 4640 seen 0.75 or 75% (accept 0.748) e.c.f. from (ii)	C1 A1
32.		
(a)	chemical energy to or K.E to heat/thermal energy/internal energy – at end	B1 B1
(b)	friction/resistive force increases	B1
(c)	(i) work = force $\times$ distance in words, number or symbols 1.2×0.08 or 0.096 or 20 or 0.208 seen 20.8 or 21	C1 C1 A1
	(ii) power = work/time or energy/time, numerical or algebraic – may use values in (i) accept 2/0.2 or 10 W 0.48 W (ecf (i) including power of ten error in (i))	C1 A1

- 33.
- (a) (WD =) Fx **or** 1680×50 C1
84 000 J A1
- (b) friction/drag/resistance **of water/air** B1
work done against friction/resistance/drag **or** energy lost due to friction/resistance/
drag **or** energy lost as heat/internal/thermal B1
- 34.
- (a) (i) air resistance increases (as speed increases) B1
(at constant speed) becomes equal to driving force/applied force etc. B1
- (ii) driving force (forward force) larger (than air resistance/backwards force) B1
- (b) (i) ($E =$) $\frac{1}{2} mv^2$ algebraic formula C1
 $\frac{1}{2} \times 75 \times 4^2$ C1
600 J A1
- (ii) ($a =$) F/m algebraic seen or 10 (N) used as force C1
 0.13 m/s^2 A1
- (c) (i) friction (in chain/axles) or rubbing of surfaces B1
heat or thermal energy produced B1
- (ii) (efficiency = useful) energy output/energy input algebraic or numerical or 380
seen C1
0.95 or 95% A1
- (d) lower mass/weight of cycle B1
less force needed same acceleration/get up hill/to stop cycle/lift cycle
greater acceleration/easier to acc. for same force or $F = ma$ quoted
less energy/work (input) to go uphill/due to less friction/ $\frac{1}{2} mv^2$ less
less kinetic energy less stopping distance/less force to stop
more efficient/less energy wasted going uphill/less friction
less friction easier turn handlebars/higher (top) speed
less pressure (on ground) sinks less into ground M1A1